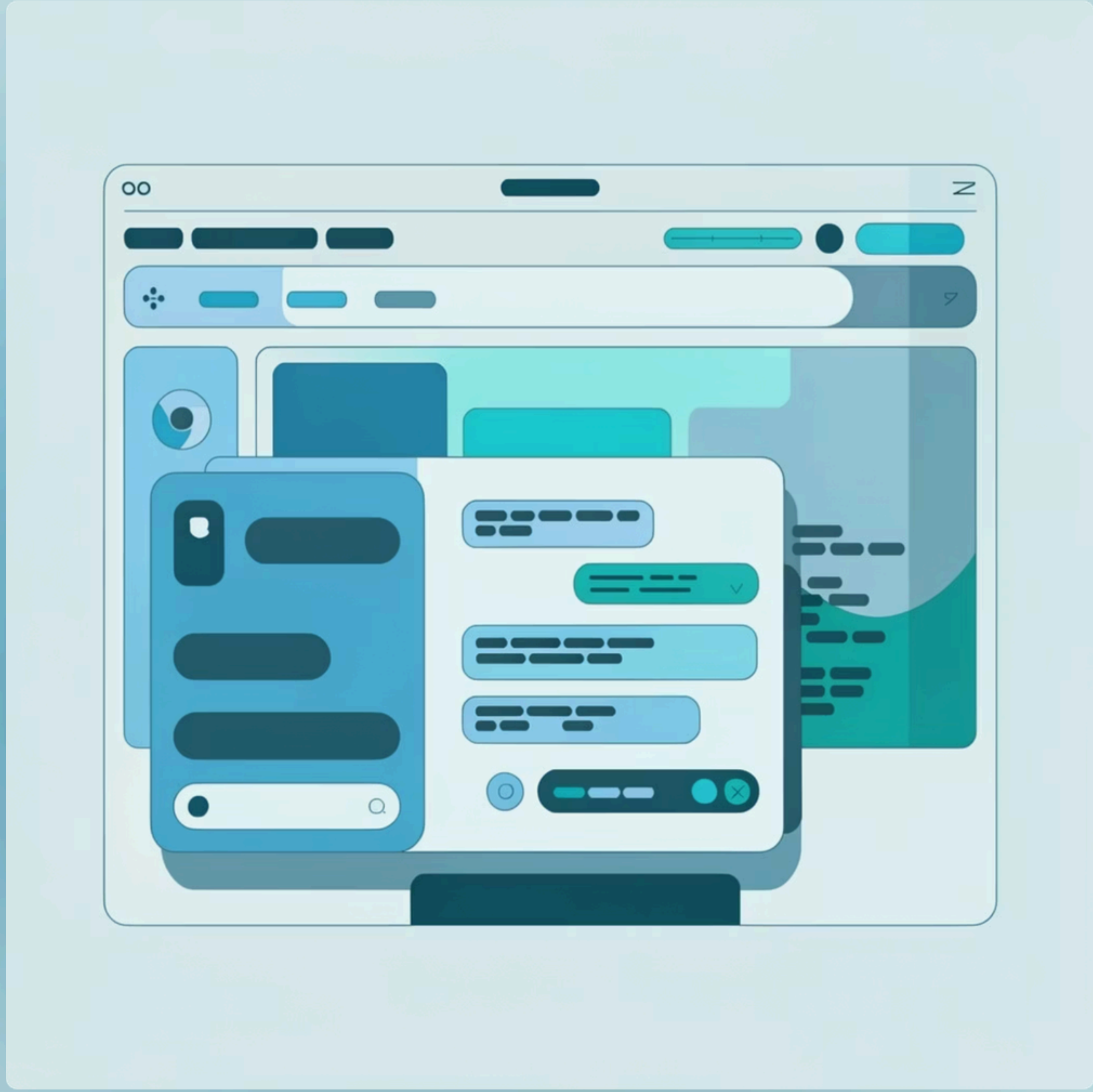
Flair Minds

ShadowLight: AI-Powered Accessibility & Navigation Assistant

An intelligent Chrome extension that transforms web accessibility through real-time AI guidance and visual assistance

Mentor – Punit Suman



Problem Statement

Challenges in Web Accessibility & Navigation

Navigation Complexity

Finding specific buttons or links in enterprise-level dashboards

Information Overload

Parsing long articles or technical documentation for key insights

Visual Accessibility

Users with color blindness or visual fatigue find it difficult to read and interact with standard web designs

Content Efficiency

Professionals need to quickly repurpose web content for other platforms (e.g., social media)

Solution Approach

ShadowLight AI Assistant - Chrome Extension

Contextual Intelligence

Real-time page scraping and analysis

Guided Interaction

Step-by-step navigation instructions

Visual Adaptation

Dynamic CSS filters to aid accessibility

Natural Language Interaction

A chatbot that "knows" what is on the user's screen

Team, Roles & Responsibilities

Project Leadership & Team Structure

Role	Name & Responsibility
Team Lead	Swarup Kusalkar - Repurpose and Summarization
Tech Lead	Sanjana Palkar - AI Engine & Navigation
Team Member	Sairaj Dhamal - DOM Scripting & Accessibility
Team Member	Samay Gangwal - UI & QA Lead
Team Member	Manasi Choudhari - QA & Documentation

Technical Details

Architecture & Tech Stack

Frameworks & Technologies

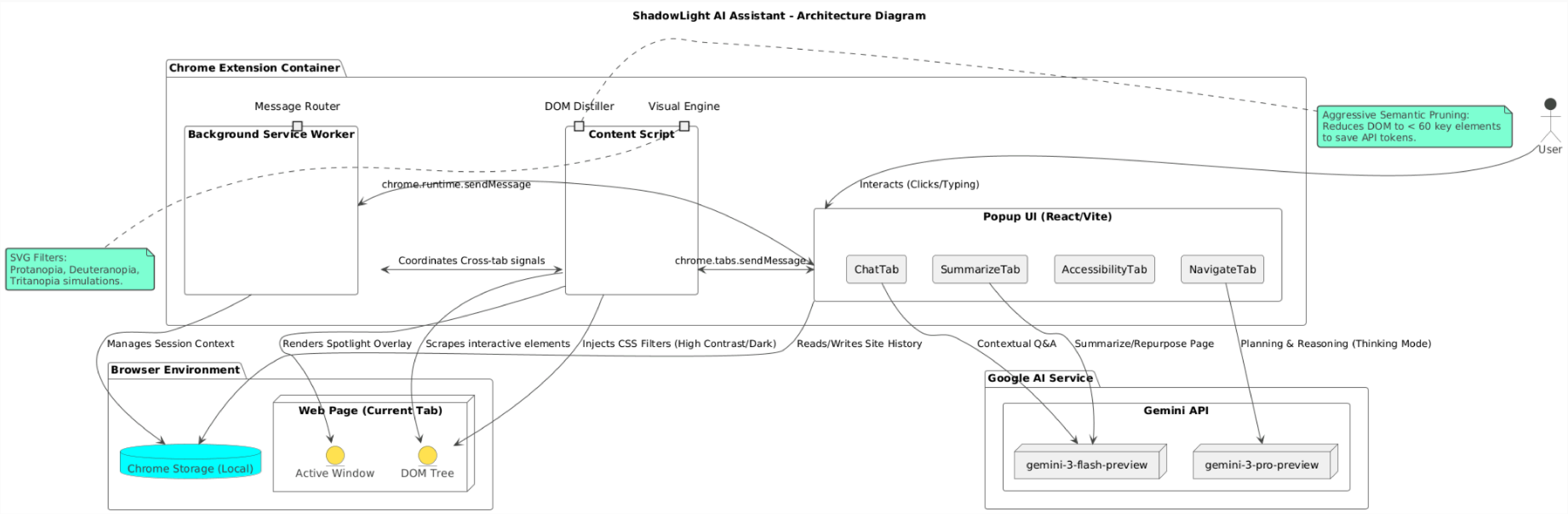
- Frameworks: React 19 (UI Layer), TypeScript (Type Safety), Vite (Build System)
- AI Engine: Google Gemini 3 (Flash for speed, Pro for reasoning)
- Architecture: Chrome Extension Manifest V3
- Key Components: Popup (App.tsx), Content Script (content.ts), Service Worker (background.ts)
- Communication: Asynchronous Message Passing via chrome.runtime.sendMessage
- Memory: Local storage caching for token-efficient cross-page context
- AI services:Antigravity IDE,GoogleAI Studio,Claude,Gemini API key.

Core Architecture

- Side Panel: React UI with Tailwind CSS
- Background Script: Service Worker for persistent logic
- Content Script: DOM interaction and element detection
- Gemini AI Service: Real-time intelligence
- Web Page DOM: Target interaction layer
- Task Management : JIRA

Architecture Overview

System Components & Data Flow



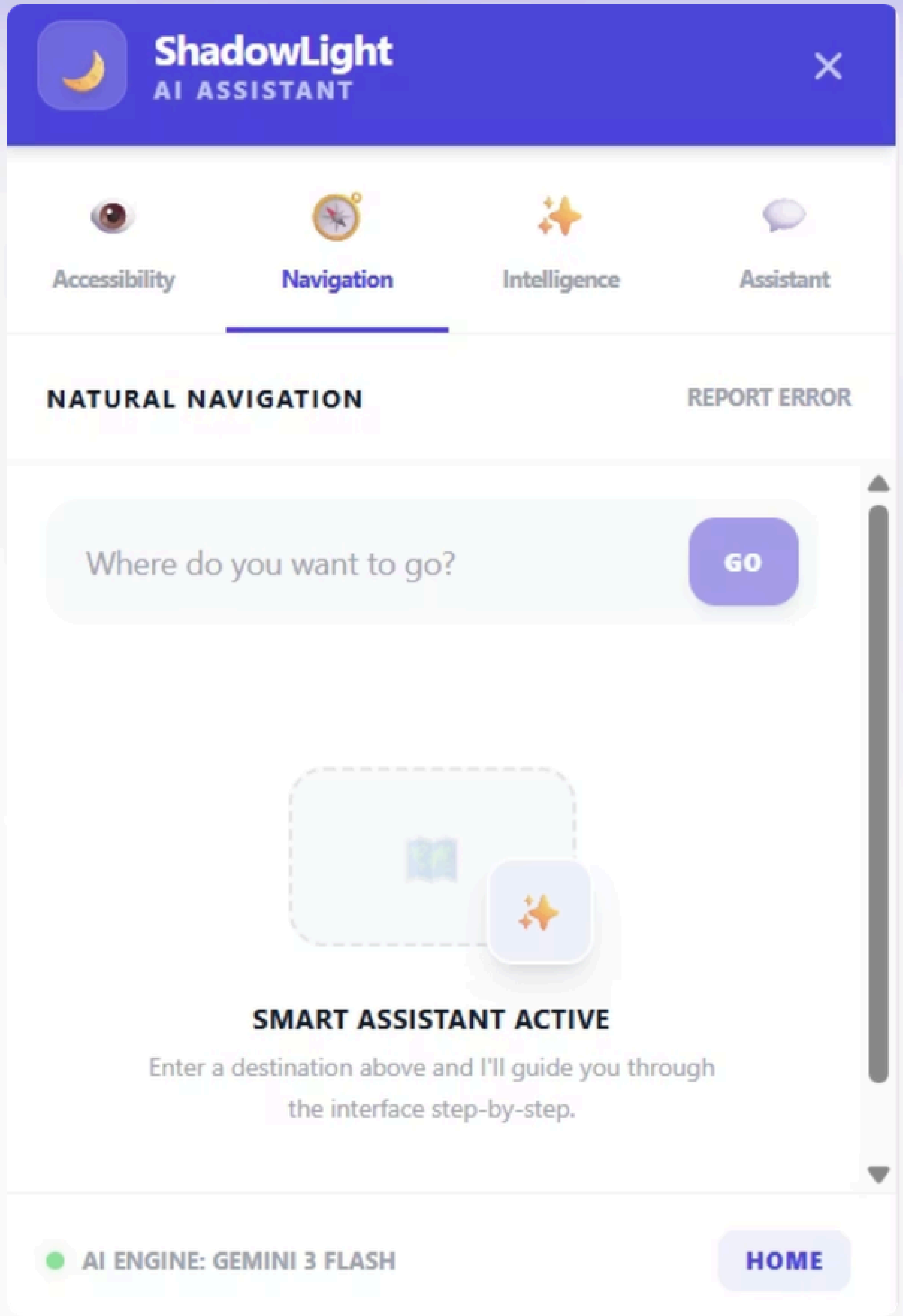
1. Navigation Module

Natural Navigation & Smart Guidance

The Navigation Module uses AI-powered guidance to help users find and interact with specific elements on any webpage through natural language commands.

Key Features:

- Natural Language Input: Simply describe what you want to do (e.g., "I want to logout")
- Smart Element Detection: AI identifies the correct button or link automatically
- Step-by-Step Guidance: Provides clear instructions for complex navigation tasks
- Sense-Plan-Act Cycle: Scrapes page, analyzes with AI, and executes actions seamlessly



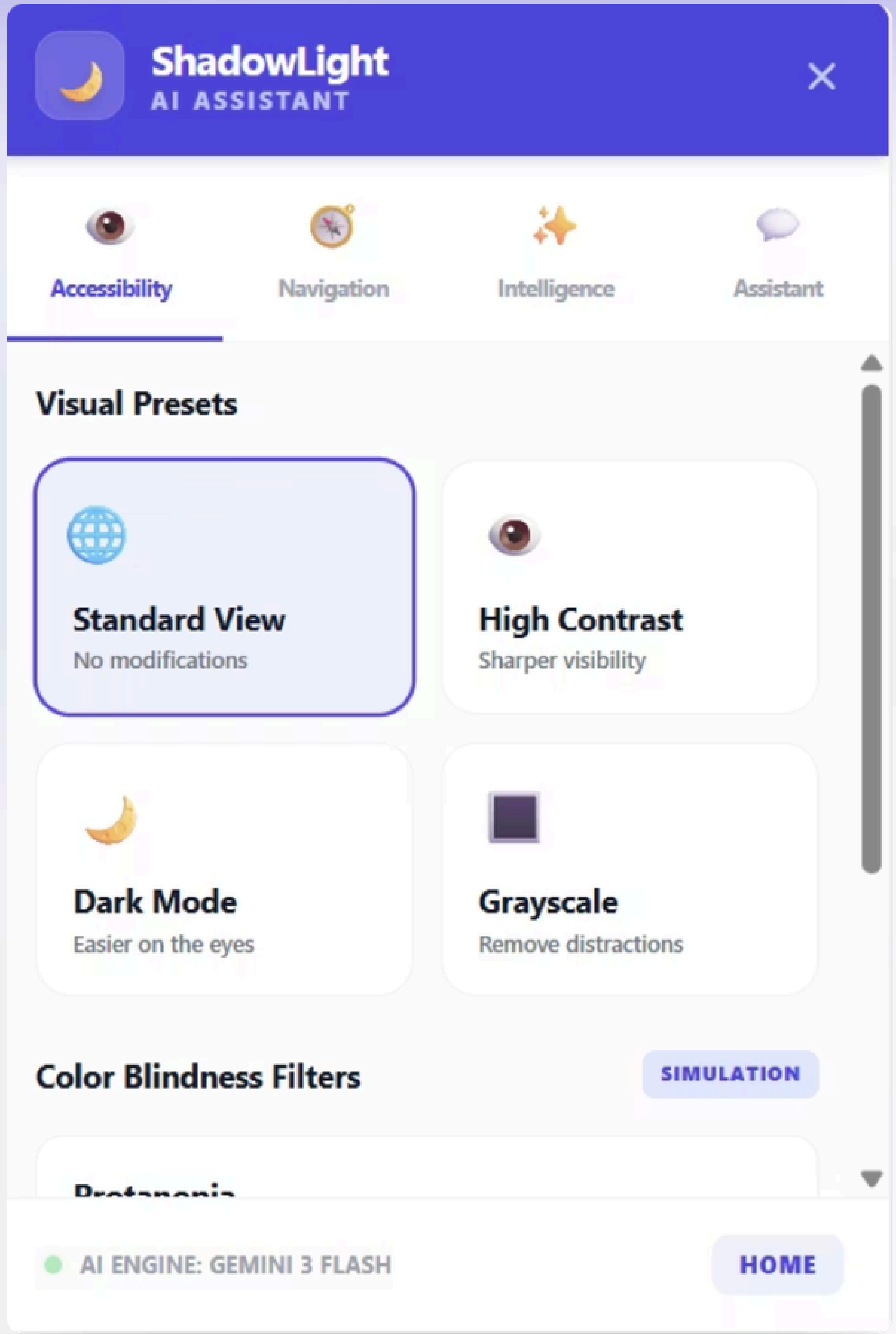
2. Accessibility Module

Visual Presets & Color Blindness Filters

The Accessibility Module provides comprehensive visual adaptation tools to make web content accessible to all users, including those with color vision deficiencies and visual fatigue.

Key Features:

- Visual Presets: One-click High Contrast, Dark Mode, and Grayscale modes
- Color Blindness Simulation: Supporting Protanopia, Deuteranopia, and Tritanopia
- Dynamic CSS Filters: Real-time application of accessibility adjustments
- User-Controlled: Toggle modes instantly in the UI with state synced to active tab



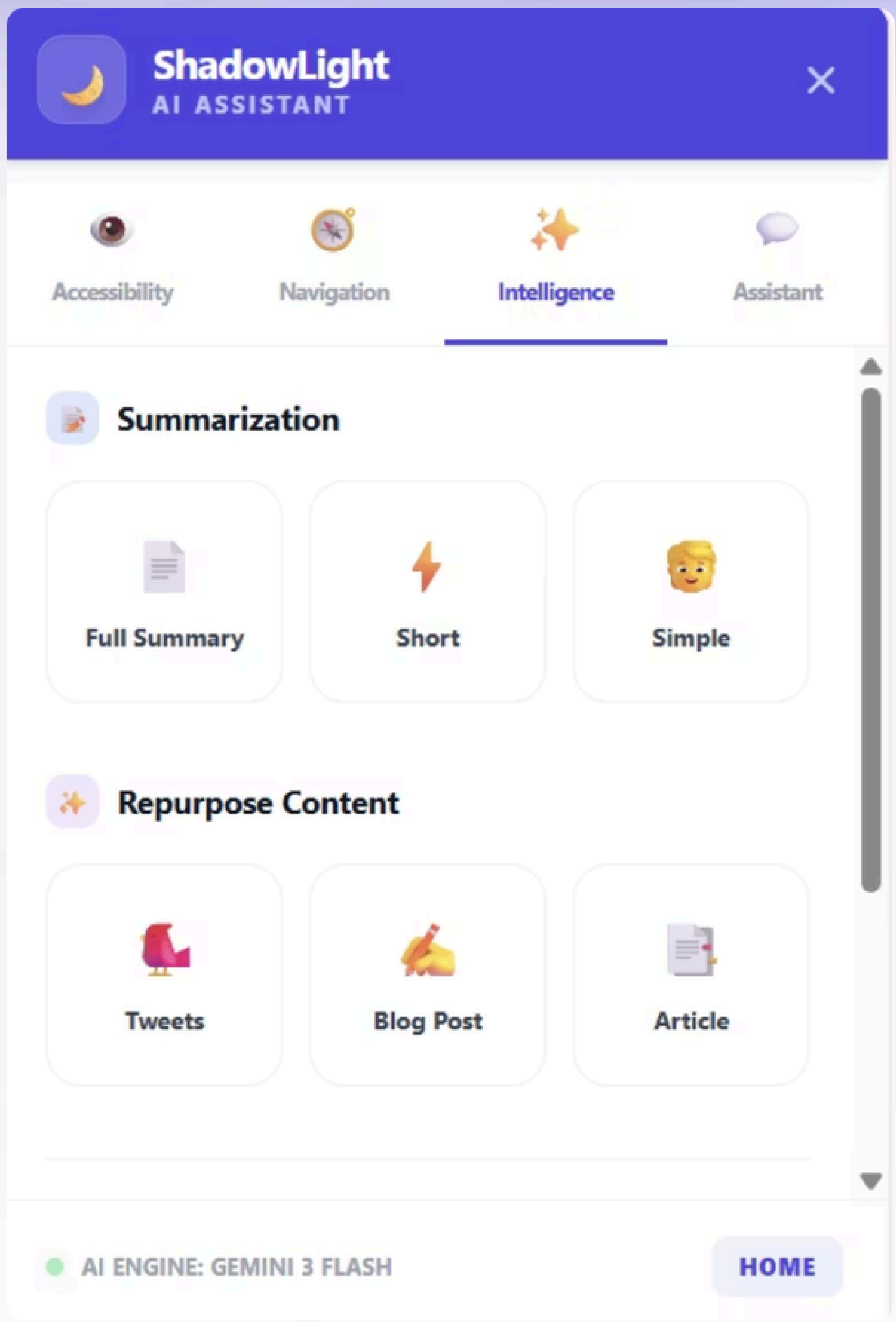
3. Intelligence Module

Smart Summarization & Content Repurposing

The Intelligence Module leverages Google Gemini 3 to understand page content and provide instant summaries or help repurpose content for different platforms.

Key Features:

- Smart Summarization: Get page summaries in ELI5 (Simple), Short, or Full formats
- Content Repurposing: Convert research into Tweets, Blog posts, or Articles instantly
- Real-Time Analysis: Powered by Google Gemini 3 (Flash for speed, Pro for reasoning)
- Context-Aware: Understands exactly what's on the current page



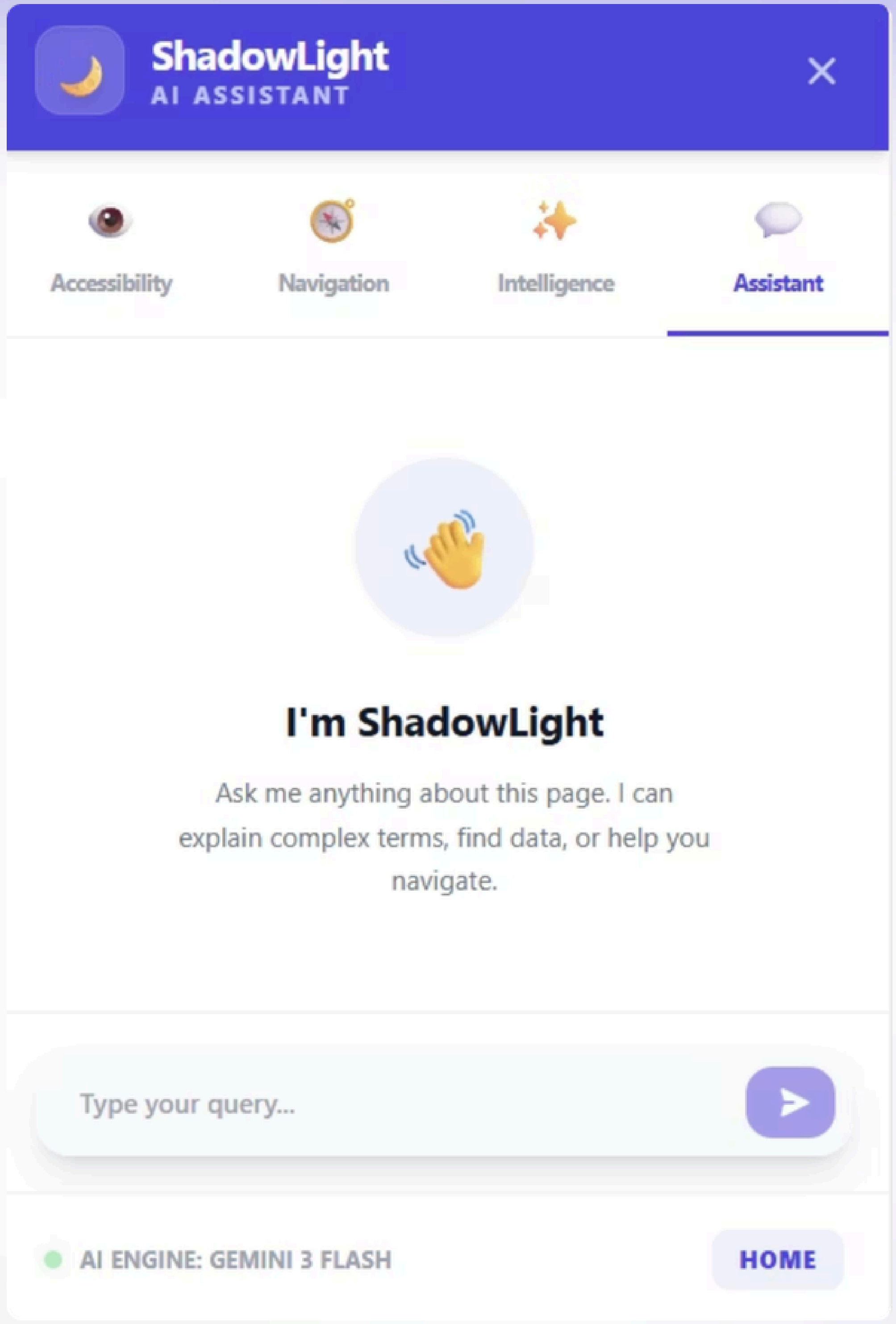
4. Assistant Module

Direct Page Chat & Conversational AI

The Assistant Module provides a conversational interface where users can ask questions about any webpage and receive intelligent, context-aware responses powered by Gemini AI.

Key Features:

- Direct Page Chat: Talk to any website as if it were a document
- Contextual Understanding: AI has full knowledge of the current page content
- Natural Conversation: Ask follow-up questions and get detailed explanations
- Multi-Purpose: Explain complex terms, find specific data, or provide navigation help
- Stateful Interaction: Maintains conversation context across multiple queries



Scope of Work

What the Project Delivered

01

Core Extension Engine

Manifest V3 setup, background/content script syncing

02

AI Integration

Secure API key management and prompt engineering for navigation and summarization

03

UI/UX Design

Creating a multi-tab sidebar with real-time UI updates

04

Accessibility Layer

Implementation of vision-simulating SVG filters and high-contrast modes

05

Features Implementation

Visual presets, color blindness simulation, page chat, smart summarization, natural navigation, content creation

06

Documentation

SRS, Technical Documentation, Architecture Flow, Timelines

07

Testing & QA

Cross-website testing, DOM selector validation, performance optimization

Key Features

ShadowLight Capabilities

Visual Presets
One-click High Contrast, Dark Mode, and Grayscale

Color Blindness Simulation
Supporting three major types of color vision deficiency (Protanopia, Deuteranopia, Tritanopia)

Direct Page Chat
Talk to any website as if it were a document

Smart Summarization
Get the gist of any page in seconds (ELI5, Short, or Full summaries)

Natural Navigation
Type "I want to logout" and let the AI find and highlight the button

Content Creation
Effortlessly repurpose research into social media drafts (Tweets, Blog posts, Articles)

Accessibility First
Automatic contrast fixes, keyboard navigation compliance, screen reader support

Advantages & Results

Key Benefits of ShadowLight

Enhanced Productivity

Reduces the cognitive load of searching for buried UI elements

Universal Accessibility

Makes the web readable for users with vision impairments

Seamless Integration

Runs in a non-intrusive side panel, keeping the main browsing space clear

Real-Time Context

Unlike standalone AI, it sees exactly what the user sees

Privacy-First Approach

No personal data stored; all processing is minimal and user-initiated

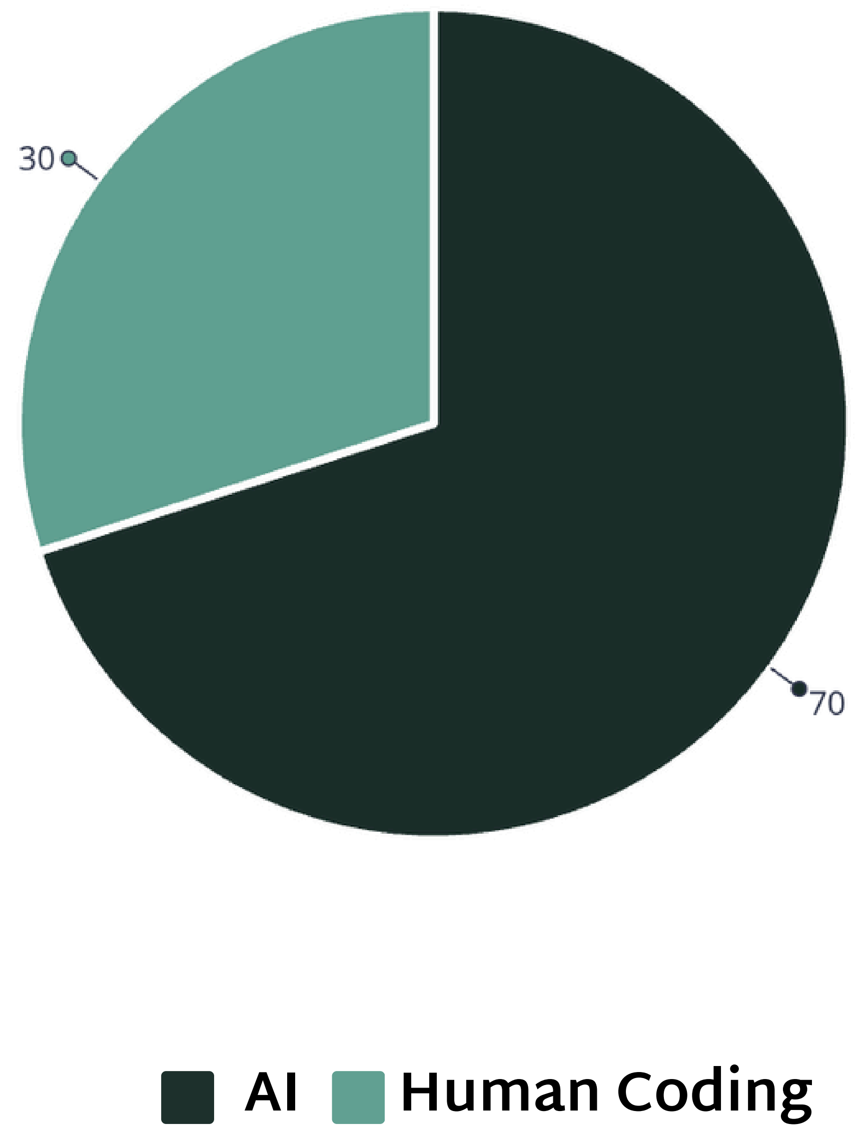
Modern & Scalable Architecture

Built using React + TypeScript + Plasmo, easy to extend and maintain

AI Impact

Accelerating Development with Intelligent Automation

Development Composition



AI Tools & Technologies Used:

- Google Gemini 3 (Core AI Engine)
- Claude (Code Generation & Analysis)
- LLaMA (Local Processing)

Time Efficiency:

- Traditional Methods: 3 weeks
- AI-Powered Approach: 1 week
- Time Saved: 66% faster development

Key Benefits:

- Rapid prototyping and iteration
- Intelligent code generation
- Automated testing and validation
- Enhanced documentation
- Reduced development cycles

Future Enhancements

Roadmap for ShadowLight Evolution

Voice Navigation

Integrating Speech-to-Text for completely hands-free web browsing

Multi-Tab Context

Allowing the AI to reference information from multiple open tabs

Personalized Accessibility

Auto-detecting page contrast issues and suggesting adjustments

Offline Modes

Using local LLMs (like Chrome Built-in AI) for basic summarization to enhance privacy

Conclusion

ShadowLight: Transforming Web Accessibility

ShadowLight represents a significant advancement in making the web more accessible and user-friendly for everyone. By combining cutting-edge AI technology with thoughtful accessibility design, we've created a tool that empowers users to navigate the digital world with confidence and ease.

Key Takeaways:

- Intelligent AI-powered assistance for web navigation and content understanding
- Comprehensive accessibility features for users with diverse needs
- Privacy-first architecture ensuring user data security
- Scalable, modern technology stack built for future growth
- Commitment to continuous improvement and user-centric innovation

The future of web accessibility is here. ShadowLight is ready to make a difference.